ParkSmart: Milestone 2 Refined Prototype

Nicholas Palmgren | CSCI 4250 Human-Computer Interaction | Fall 2026

Runnable prototype: https://github.com/chickencell/ParkSmart/tree/main/m2-prototype (open index.html and click the screens)
Video demonstration: [paste YuJa link here]

The prototype

ParkSmart helps a student find an open parking lot near the building they are going to. The refined prototype is a clickable set of hand-drawn style screens: tapping a sketch moves to the next screen, the way paper screens are swapped in a paper-prototype test. Students can scan the campus map (lots colored green / yellow / red with open-spot counts), tap a lot for a quick preview, or pick a building to get the closest lots that still have space, check a lot's details, and hand off to their maps app for directions. Availability is simulated.

Design changes since Milestone 1

- **Lot preview card.** Tapping a map marker now opens a small card over the map (open spots, permit, walk time) instead of jumping to a new screen, so users can compare lots without losing the map.
- **Recent buildings.** Find a Building lists the last places searched, since students go to the same few buildings; picking from a list is easier than remembering and typing a name.
- **"All lots full" screen.** Milestone 1 had no answer when nothing nearby was open. Now the app says so plainly and suggests the closest lot that has space.
- **Fill bars and "Updated 2 min ago."** Best Lots cards show a fill bar for quick comparison, and a timestamp tells users how fresh the numbers are.
- **Back to map.** Lot Info has a button to return home in one tap.

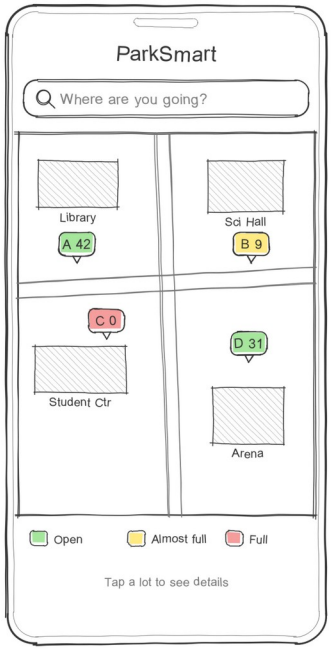
Usage scenarios (draft)

1. Running late for class. Maya commutes and has a 10:00 class in Science Hall. At 9:50 she pulls into campus, opens ParkSmart, taps Science Hall under Recent, and sees Lot A has 42 open spots and is a 4-minute walk. She taps Get Directions and parks without circling full lots.

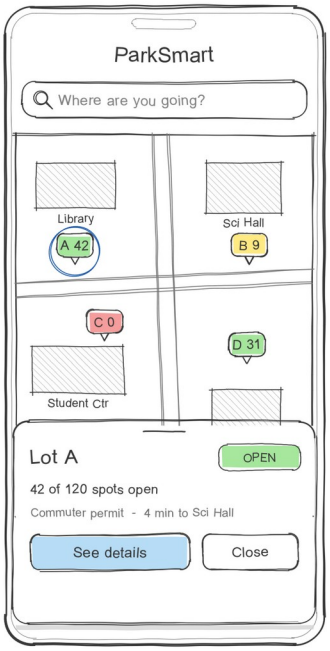
2. Checking before leaving home. Jordan opens the Campus Map before leaving his apartment. Lot C is red (full), so he taps the green Lot A marker, sees 42 open spots in the preview card, and heads there.

3. Event night. Sam drives to a game at the Arena. The app shows that all lots near the Arena are full and suggests Lot A, a 9-minute walk, so Sam goes straight there instead of searching.

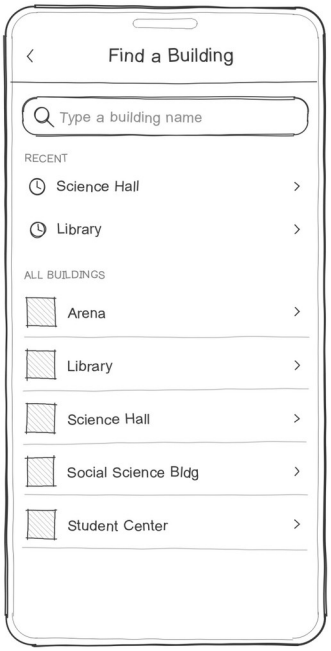
Prototype screens



1. Campus Map



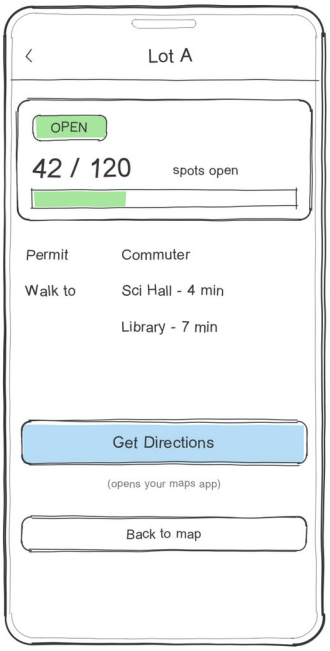
2. Lot preview (new)



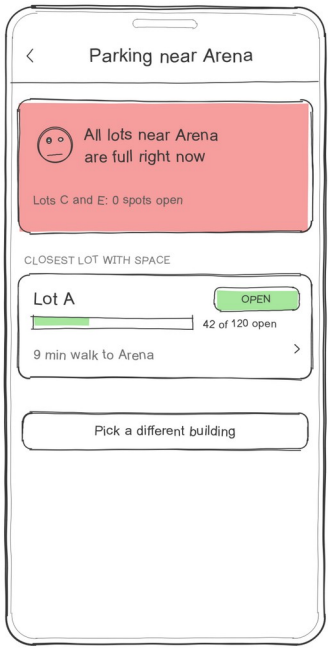
3. Find a Building



4. Best Lots



5. Lot Info



6. All lots full (new)

Paths through the prototype: Scenario 1: search bar → Science Hall → Lot A → Get Directions. Scenario 2: tap a lot marker → See details. Scenario 3: search bar → Arena → Lot A.

AI disclosure: I used Claude (Anthropic) to help draw the sketches, build the clickable prototype and draft this writeup.